

Addition of angular momenta[★]

OUTLINE

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In the previous chapter, we analyzed systems with a single angular momentum, $\vec{L}$, which operates within a Hilbert space of dimension $2l + 1$. However, in nature, we often encounter systems with multiple angular momenta, such as orbital momentum $\vec{L}$ and spin $\vec{S}$, which interact through various types of interactions represented by their respective Hamiltonian formulations. To address these systems, this chapter begins by exploring Hamiltonians that describe these interactions. Following this, we explore the addition of angular momenta and the formation of commuting ob-

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servables, which are essential for understanding how angular momenta combine. The discussion then expands to the Hilbert space associated with multiple angular momenta and the corresponding Clebsch–Gordan coefficients, applicable to up to N spins. Finally, the chapter connects the theoretical framework to practical applications, such as the calculation of magnetic moments (requiring the addition of orbital and spin angular momenta), and the total angular momentum of atoms (analyzed through Hund’s Rules).

Objective The goal of this chapter is to extend the study of angular momentum to systems involving multiple angular momenta. It aims to provide a comprehensive understanding of how angular momenta interact and sum up, forming their bases of commuting observables. The chapter introduces the mathematical tools necessary to describe these systems, including the expansion of the Hilbert space and the derivation of change of basis matrices using Clebsch–Gordan coefficients. This chapter also explores applications, such as magnetic moments and the coupling of atomic angular momentum.

7.1 Interactions between angular momenta

In this section, we introduce Hamiltonians that describe interactions between angular momenta. The purpose of this presentation is to provide physical insights that motivate the addition of angular momenta. For a more in-depth discussion on the physical mechanisms underlying these interactions, we refer the reader to reference [17].

Heisenberg: This isotropic interaction originates from the superposition of the atomic orbitals of two neighboring ions (which, for simplicity, we are considering a spin-only contribution). This Hamiltonian can be written as follows:

$$\hat{\mathcal{H}} = -J \vec{\hat{S}}_1 \cdot \vec{\hat{S}}_2. \quad (7.1)$$

For $J > 0$ the spins are ordered parallel, while for $J < 0$ the spins are ordered antiparallel. Note that J regulates the intensity of the interaction.

Ising: This is a particular case of the Heisenberg Hamiltonian and has an anisotropic character, where the spins are along the z axis. The mathematical formulation of the Hamiltonian is given by:

$$\hat{\mathcal{H}} = -J \hat{S}_{1z} \hat{S}_{2z}. \quad (7.2)$$

XY: Similarly to the previous case, this interaction has an anisotropic character where the spins are on the xy plane, with the Hamiltonian formulation given by:

$$\hat{H} = -J(\hat{S}_{1x}\hat{S}_{2x} + \hat{S}_{1y}\hat{S}_{2y}). \quad (7.3)$$

Antisymmetric: This interaction points the angular momenta orthogonally to each other. For simplicity, we are considering only the spin contribution and thus this Hamiltonian can be written as:

$$\hat{H} = \vec{d} \cdot (\hat{S}_1 \times \hat{S}_2). \quad (7.4)$$

Note that the parameter that regulates the intensity of the interaction is the $\vec{d}$ vector. This term is also known as the Dzialoshinsky–Moriya interaction.

Spin-orbit: The orbital angular momentum $\vec{L}$ of an electron can interact with its own spin angular momentum $\vec{S}$. The parameter ζ gives the intensity of this interaction and will be further discussed in Application 7.B and Subsection 13.4.2. The corresponding Hamiltonian can be written as:

$$\hat{H} = \zeta \vec{S} \cdot \vec{L}. \quad (7.5)$$

Local magnetocrystalline anisotropy: This Hamiltonian has an anisotropic character and originates from the spin-orbit interaction. In few words, this is the interaction of a spin with the crystal lattice. This Hamiltonian is described as:

$$\hat{H} = D \left(\hat{S}_z^2 - \frac{1}{3} \hat{S}^2 \right) + E \left(\hat{S}_x^2 - \hat{S}_y^2 \right). \quad (7.6)$$

The parameters D (axial) and E (rhombic) regulate the intensity of this interaction.

Zeeman: This case represents the interaction between angular momenta (spin and/or orbital) with a magnetic field. The Hamiltonian can therefore be written as:

$$\hat{H} = \mu_B \left(2\vec{S} + \vec{L} \right) \cdot \vec{B}. \quad (7.7)$$



In real units, all Hamiltonians above have a pre-factor $1/\hbar^2$ (except for the Zeeman Hamiltonian, whose pre-factor is $1/\hbar$). This pre-factor exists due to the units of angular momentum ($\hbar$) – considering that the parameters that regulate the intensity of interactions (J , ζ , etc...) have units of energy. However, for simplicity, we express the Hamiltonians without these pre-factors and with the respective dimensionless angular momenta.

7.2 Commutation relations

As we have seen above, there are several possibilities of interaction between angular momenta (and many others that were not mentioned) and the addition of angular momentum depends on the system under consideration.

Objective Our goal for this section is to describe the set of commuting observables representing the physical system under study.

To begin, let's consider only two angular momenta: $\vec{\hat{L}}_1$ and $\vec{\hat{L}}_2$, where each has its own Hilbert subspace, and these are independent. In this way, we propose that the components of the angular momentum operators belonging to different subspaces commute with each other, such that:

$$[\hat{L}_{1u}, \hat{L}_{2v}] = 0, \quad (7.8)$$

where $(u, v) = (x, y, z)$. The commutation relation above implies:

$$[\hat{L}_1^2, \hat{L}_{2v}] = 0, \quad [\hat{L}_2^2, \hat{L}_{1v}] = 0 \quad \text{and} \quad [\hat{L}_1^2, \hat{L}_2^2] = 0. \quad (7.9)$$

Furthermore, each subspace obeys the commutation rule of angular momentum (see Eq. (6.4)):

$$[\hat{L}_{1u}, \hat{L}_{1v}] = i\hbar\epsilon_{uvw}\hat{L}_{1w} \quad \text{and} \quad [\hat{L}_{2u}, \hat{L}_{2v}] = i\hbar\epsilon_{uvw}\hat{L}_{2w} \quad (7.10)$$

and, consequently, the entire algebra of angular momentum:

$$\hat{L}_{1z}|l_1, m_1\rangle = m_1\hbar|l_1, m_1\rangle \quad \text{and} \quad \hat{L}_1^2|l_1, m_1\rangle = l_1(l_1 + 1)\hbar^2|l_1, m_1\rangle, \quad (7.11)$$

$$\hat{L}_{2z}|l_2, m_2\rangle = m_2\hbar|l_2, m_2\rangle \quad \text{and} \quad \hat{L}_2^2|l_2, m_2\rangle = l_2(l_2 + 1)\hbar^2|l_2, m_2\rangle. \quad (7.12)$$

Therefore:

$$\{\hat{L}_1^2, \hat{L}_2^2, \hat{L}_{1z}, \hat{L}_{2z}\} \quad (7.13)$$

forms a set of commuting observables represented by the vector $|l_1, m_1\rangle \otimes |l_2, m_2\rangle = |l_1, l_2, m_1, m_2\rangle$. This basis is known as *local*.

However, for certain Hamiltonians, the local basis may not diagonalize the operator, necessitating the use of an alternative basis. Let's now focus on the total angular momentum of the system, that is:

$$\vec{\hat{L}} = \vec{\hat{L}}_1 + \vec{\hat{L}}_2 \quad (7.14)$$

where

$$\hat{L}^2 = (\vec{\hat{L}}_1 + \vec{\hat{L}}_2)^2 = \hat{L}_1^2 + \hat{L}_2^2 + 2\vec{\hat{L}}_1 \cdot \vec{\hat{L}}_2, \quad (7.15)$$

$$\hat{L}_x = \hat{L}_{1x} + \hat{L}_{2x}, \quad (7.16)$$

$$\hat{L}_y = \hat{L}_{1y} + \hat{L}_{2y}, \quad (7.17)$$

$$\hat{L}_z = \hat{L}_{1z} + \hat{L}_{2z}, \quad (7.18)$$

$$\hat{L}_{\pm} = \hat{L}_{1\pm} + \hat{L}_{2\pm}. \quad (7.19)$$

Since we aim to establish a new set of commuting observables that incorporates the total angular momentum, it is necessary to introduce additional commutation relations:

$$[\hat{L}^2, \hat{L}_1^2] = [\hat{L}^2, \hat{L}_2^2] = 0. \quad (7.20)$$

The details of these commutation relations are provided in Box 7.1. At this stage, we have identified three commuting observables. However, the local basis consists of four observables, meaning one additional observable is needed. To address this, we examine another commutation relation, which is demonstrated in detail in Box 7.2:

$$[\hat{L}^2, \hat{L}_z] = 0. \quad (7.21)$$

We therefore have a new set of commuting observables, with the same number of quantum numbers as the local basis. The new set is:

$$\{\hat{L}_1^2, \hat{L}_2^2, \hat{L}^2, \hat{L}_z\} \quad (7.22)$$

represented by vector $|l_1, l_2, l, m\rangle$. This basis is known as *coupled*.

How do the components of the total angular momentum commute with one another? To begin, we can examine the commutation relation between two components:

$$\begin{aligned} [\hat{L}_x, \hat{L}_y] &= [\hat{L}_{1x} + \hat{L}_{2x}, \hat{L}_{1y} + \hat{L}_{2y}] \\ &= [\hat{L}_{1x}, \hat{L}_{1y}] + [\hat{L}_{2x}, \hat{L}_{2y}] \\ &= (i\hbar)\hat{L}_{1z} + (i\hbar)\hat{L}_{2z} \\ &= i\hbar\hat{L}_z. \end{aligned} \quad (7.23)$$

The other components are left to the reader verify. We can generalize the above result as follows:

$$[\hat{L}_u, \hat{L}_v] = i\hbar\epsilon_{uvw}\hat{L}_w \quad (7.24)$$

that is, the components of the total angular momentum follow the same commutation relation as a single angular momentum. Consequently, the total angular momentum follows the algebra discussed in the previous

chapter and therefore we know its eigenvalues and vectors (see Eqs. (6.48), (6.49) and (6.50)):

$$\hat{L}_z|l, m\rangle = m\hbar|l, m\rangle \quad \text{and} \quad \hat{L}^2|l, m\rangle = l(l+1)\hbar^2|l, m\rangle \quad (7.25)$$

where

$$-l \leq m \leq +l. \quad (7.26)$$

What are the possible values that l can take? Obviously, l must be between l_{min} and l_{max} . We know that $m_{1max} = l_1$ and $m_{2max} = l_2$, then $m_{max} = l_{max}$. However, $m_{max} = m_{1max} + m_{2max}$ and therefore $l_{max} = l_1 + l_2$. What about l_{min} ? The dimension of the Hilbert space must be the same whether we consider the local or coupled basis, that is:

$$(2l_1 + 1)(2l_2 + 1) = \sum_{l=l_{min}}^{l_{max}} (2l + 1). \quad (7.27)$$

The summation above has an analytical solution,¹ and thus:

$$(2l_1 + 1)(2l_2 + 1) = (1 + l_{max} + l_{min})(1 + l_{max} - l_{min}). \quad (7.29)$$

However $l_{max} = l_1 + l_2$ and then:

$$(2l_1 + 1)(2l_2 + 1) = [(l_1 + l_2 + l_{min}) + 1][(l_1 + l_2 - l_{min}) + 1]. \quad (7.30)$$

By comparing both sides of the above equation, we obtain $l_{min} = l_2 - l_1$ and $l_{min} = l_1 - l_2$. Therefore, there is only one solution: $l_{min} = |l_1 - l_2|$. Thus, the possible values of l are:

$$|l_1 - l_2| \leq l \leq l_1 + l_2 \quad (7.31)$$

Each value of l corresponds to a multiplet of the system.

There is also another important constraint. What is the relation between the values of m_1 and m_2 and the values of m ? Let's consider the following matrix element:

$$\langle l_1, l_2, m_1, m_2 | \hat{L}_z | l_1, l_2, l, m \rangle = m\hbar \langle l_1, l_2, m_1, m_2 | l_1, l_2, l, m \rangle. \quad (7.32)$$

Nevertheless, using Eq. (7.18), the above equation can be rewritten as follows:

$$\langle l_1, l_2, m_1, m_2 | \hat{L}_{1z} + \hat{L}_{2z} | l_1, l_2, l, m \rangle = (m_1 + m_2)\hbar \langle l_1, l_2, m_1, m_2 | l_1, l_2, l, m \rangle. \quad (7.33)$$

1

$$\sum_{k=a}^b (2k + 1) = (1 + b + a)(1 + b - a). \quad (7.28)$$

The left side of Eqs. (7.32) and (7.33) is equal and therefore we conclude that:

$$m = m_1 + m_2 \quad (7.34)$$

To think This is the necessary condition, but not sufficient, for the matrix element $\langle m_1, m_2 | l, m \rangle$ to be nonzero.

Box 7.1

Detailed demonstration of commutations $[\hat{L}^2, \hat{L}_1^2] = [\hat{L}^2, \hat{L}_2^2] = 0$

$$\begin{aligned} [\hat{L}^2, \hat{L}_1^2] &= [\hat{L}_1^2 + \hat{L}_2^2 + 2\vec{\hat{L}}_1 \cdot \vec{\hat{L}}_2, \hat{L}_1^2] \\ &= 2[\vec{\hat{L}}_1 \cdot \vec{\hat{L}}_2, \hat{L}_1^2], \end{aligned} \quad (7.35)$$

where the subspaces are considered to be independent and the operators from different subspaces always commute. To determine $[\hat{L}^2, \hat{L}_1^2]$ we need more algebra:

$$\begin{aligned} [\vec{\hat{L}}_1 \cdot \vec{\hat{L}}_2, \hat{L}_1^2] &= [\hat{L}_{1x}\hat{L}_{2x} + \hat{L}_{1y}\hat{L}_{2y} + \hat{L}_{1z}\hat{L}_{2z}, \hat{L}_{1x}^2 + \hat{L}_{1y}^2 + \hat{L}_{1z}^2] \\ &= [\hat{L}_{1y}, \hat{L}_{1x}^2]\hat{L}_{2y} + [\hat{L}_{1z}, \hat{L}_{1x}^2]\hat{L}_{2z} + [\hat{L}_{1x}, \hat{L}_{1y}^2]\hat{L}_{2x} \\ &\quad + [\hat{L}_{1z}, \hat{L}_{1y}^2]\hat{L}_{2z} + [\hat{L}_{1x}, \hat{L}_{1z}^2]\hat{L}_{2x} + [\hat{L}_{1y}, \hat{L}_{1z}^2]\hat{L}_{2y}. \end{aligned} \quad (7.36)$$

However, it is easy to show that:

$$[\hat{L}_u, \hat{L}_v^2] = (i\hbar)\epsilon_{uvw}\{\hat{L}_v, \hat{L}_w\}, \quad (7.37)$$

where $\{\hat{A}, \hat{B}\} = \hat{A}\hat{B} + \hat{B}\hat{A}$ is the anti-commutator. Thus, we can verify that:

$$\begin{aligned} [\vec{\hat{L}}_1 \cdot \vec{\hat{L}}_2, \hat{L}_1^2] &= (i\hbar)[-\{\hat{L}_{1x}, \hat{L}_{1z}\}\hat{L}_{2y} + \{\hat{L}_{1x}, \hat{L}_{1y}\}\hat{L}_{2z} + \{\hat{L}_{1y}, \hat{L}_{1z}\}\hat{L}_{2x} \\ &\quad - \{\hat{L}_{1y}, \hat{L}_{1x}\}\hat{L}_{2z} - \{\hat{L}_{1z}, \hat{L}_{1y}\}\hat{L}_{2x} + \{\hat{L}_{1z}, \hat{L}_{1x}\}\hat{L}_{2y}] \\ &= 0 \end{aligned} \quad (7.38)$$

and, consequently:

$$[\hat{L}^2, \hat{L}_1^2] = [\hat{L}^2, \hat{L}_2^2] = 0. \quad (7.39)$$

Box 7.2 Detailed demonstration of commutation $[\hat{L}^2, \hat{L}_z] = 0$

$$\begin{aligned}
[\hat{L}^2, \hat{L}_z] &= [\hat{L}_1^2 + \hat{L}_2^2 + 2\vec{\hat{L}}_1 \cdot \vec{\hat{L}}_2, \hat{L}_{1z} + \hat{L}_{2z}] \\
&= 2[\vec{\hat{L}}_1 \cdot \vec{\hat{L}}_2, \hat{L}_{1z} + \hat{L}_{2z}] \\
&= 2[\hat{L}_{1x}\hat{L}_{2x} + \hat{L}_{1y}\hat{L}_{2y} + \hat{L}_{1z}\hat{L}_{2z}, \hat{L}_{1z} + \hat{L}_{2z}] \\
&= 2\left\{[\hat{L}_{1x}, \hat{L}_{1z}]\hat{L}_{2x} + [\hat{L}_{1y}, \hat{L}_{1z}]\hat{L}_{2y} + \hat{L}_{1x}[\hat{L}_{2x}, \hat{L}_{2z}] \right. \\
&\quad \left. + \hat{L}_{1y}[\hat{L}_{2y}, \hat{L}_{2z}]\right\} \\
&= 2(i\hbar)\left\{-\hat{L}_{1y}\hat{L}_{2x} + \hat{L}_{1x}\hat{L}_{2y} - \hat{L}_{1x}\hat{L}_{2y} + \hat{L}_{1y}\hat{L}_{2x}\right\} \\
&= 0.
\end{aligned} \tag{7.40}$$

■ Example 7.1

Let's consider two angular momenta: $l_1 = l_2 = 1/2$ and determine the local and coupled basis vector. The local basis is given by $|l_1, l_2, m_1, m_2\rangle$, while the coupled basis is given by $|l_1, l_2, l, m\rangle$. For the local basis, we have to write all possible combinations of m_1 and m_2 , therefore being:

$$\text{local basis } (|m_1, m_2\rangle) : \left\{ \begin{array}{l} |+, +\rangle \\ |+, -\rangle \\ |-, +\rangle \\ |-, -\rangle \end{array} \right\} \tag{7.41}$$

Note that the local basis was written in a short notation, with only the values of m_1 and m_2 , since l_1 and l_2 are constants. In addition, we are using '+' and '-' as $+1/2$ and $-1/2$, respectively.

To construct the coupled basis, we first determine the possible values of l , which range from $|l_1 - l_2|$ to $l_1 + l_2$ (see Eq. (7.31)). In this case, the possible values are $l = \{0, 1\}$. Using Eq. (7.26), we know that for each multiplet, the values of m span from $-l$ to $+l$. Therefore, the coupled basis can be expressed as:

$$\text{coupled basis } (|l, m\rangle) : \left\{ \begin{array}{l} |1, 1\rangle \\ |1, 0\rangle \\ |1, -1\rangle \\ |0, 0\rangle \end{array} \right\} \tag{7.42}$$

It is important to note that both bases preserve the dimension of the Hilbert space. For the local basis, the dimension is given by $\prod_{i=1}^2 (2l_i + 1) = (2l_1 + 1)(2l_2 + 1) = 4$. Similarly, for the coupled basis, the dimension is determined by $\sum_{l=0}^1 (2l + 1) = 4$. ■

7.3 Vectors in the coupled basis

In this section, we explain the process of constructing the vectors for a system with N angular momenta. In Example 7.1, we demonstrated this process for the simpler case of two angular momenta. To extend this to N angular momenta, the approach involves iterative summation: first, combine two angular momenta. Next, add a third angular momentum to the result, then a fourth, and continue this process until all N angular momenta have been incorporated. It is crucial to ensure that the total angular momentum resulting from each step of the summation adheres to the constraint specified in Eq. (7.31).

Objective The goal of this subsection is to establish a methodology to build the coupled basis vector, which can be sequential or non-sequential.

7.3.1 Sequentially coupled basis

In this case, the angular momenta are added sequentially: first, the first angular momentum is combined with the second. Next, the resulting total is combined with the third angular momentum, followed by the fourth, and so on, until the N -th angular momentum is included. The best way to illustrate this process is through an example.

■ Example 7.2

Let's consider four angular momenta: $l_1 = l_2 = l_3 = l_4 = 1/2$. As discussed above, we should couple them as follows:

$$\vec{\tilde{L}}_{12} = \vec{\tilde{L}}_1 + \vec{\tilde{L}}_2, \quad (7.43)$$

$$\vec{\tilde{L}}_{13} = \vec{\tilde{L}}_{12} + \vec{\tilde{L}}_3, \quad (7.44)$$

$$\vec{\tilde{L}} = \vec{\tilde{L}}_{14} = \vec{\tilde{L}}_{13} + \vec{\tilde{L}}_4. \quad (7.45)$$

From Eq. (7.31), we have the following constraints:

$$|l_1 - l_2| \leq l_{12} \leq l_1 + l_2, \quad (7.46)$$

$$|l_{12} - l_3| \leq l_{13} \leq l_{12} + l_3, \quad (7.47)$$

$$|l_{13} - l_4| \leq l \leq l_{13} + l_4, \quad (7.48)$$

which allow us to build the sequential coupled basis. First, we check the possible values for l_{12} , being: 1 and 0. Then, to build l_{13} , we add l_3 and each possible values of l_{12} . For $l_{12} = 1$, we have two possible values of l_{13} : $3/2$ and $1/2$; and for $l_{12} = 0$, there is only one possible value of l_{13} : $1/2$. To build l_{14} , we follow the same procedure. Fig. 7.1

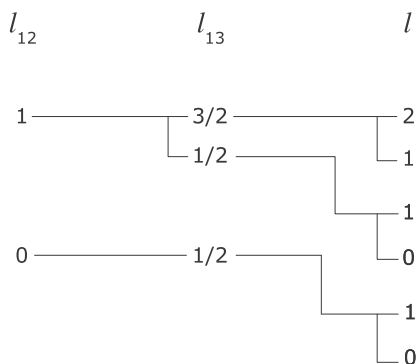


FIGURE 7.1 Addition of four $1/2$ angular momenta to build the sequential coupled basis.

summarizes this idea. Now, to build the basis, we follow all the possible lines in that figure. Thus, the coupled basis is $|l_{12}, l_{13}, l, m\rangle$, where m is the projection of the multiplet l on the quantization axis z . Note that this is a simplified notation, where the constant quantum numbers l_1 , l_2 , l_3 , and l_4 are omitted. Finally, the vectors are given by:

$$\begin{aligned}
 &|l_{12}, l_{13}, l, m\rangle \\
 &|1, 3/2, 2, m\rangle \\
 &|1, 3/2, 1, m\rangle \\
 &|1, 1/2, 1, m\rangle \\
 &|1, 1/2, 0, m\rangle \\
 &|0, 1/2, 1, m\rangle \\
 &|0, 1/2, 0, m\rangle
 \end{aligned} \tag{7.49}$$

Note that in the local basis we have four quantum numbers (m_1 , m_2 , m_3 , and m_4), one for each angular momentum. Obviously, in the coupled basis, we have the same number of quantum numbers. Furthermore, the two bases must preserve the dimension of the Hilbert space. The local basis has dimension $\prod_{i=1}^4 (2l_i + 1) = (2l_1 + 1)(2l_2 + 1)(2l_3 + 1)(2l_4 + 1) = 16$. The coupled basis has dimension $\sum_l g_l (2l + 1)$, where l represents each multiplet of the system, and g_l its multiplicity. We then have for the coupled basis: one quintuplet ($l = 2$), three triplets ($l = 1$), and two singlets ($l = 0$), resulting in a Hilbert space with dimension $(1 \times 5) + (3 \times 3) + (2 \times 1) = 16$. ■

7.3.2 Non-sequentially coupled basis

The angular momenta do not need to be added sequentially, as long as they are always added in pairs – there is freedom of choice, which depends on the system studied. For a system with three angular momenta, for example, we can first combine the first and third angular momenta, and then add the second angular momentum to the resulting total. Similarly, we can first combine the second and third angular momenta, and then add the resulting sum to the first angular momentum. The number of possibilities obviously increases with the number N of angular momenta in the system. As before, an example is the best way to explain how to build a non-sequential coupled basis.

■ Example 7.3

Let's continue with the example of four angular momenta $l_1 = l_2 = l_3 = l_4 = 1/2$ and add them together as follows:

$$\begin{aligned}\vec{\tilde{L}}_{12} &= \vec{\tilde{L}}_1 + \vec{\tilde{L}}_2, \\ \vec{\tilde{L}}_{34} &= \vec{\tilde{L}}_3 + \vec{\tilde{L}}_4, \\ \vec{\tilde{L}} &= \vec{\tilde{L}}_{12} + \vec{\tilde{L}}_{34}.\end{aligned}\tag{7.50}$$

The resultant vector sum must satisfy the conditions outlined in Eq. (7.31). To construct the eigenvectors, we follow the same procedure described in the previous case. Fig. 7.2 illustrates the diagram for the addition of these vectors. Finally, the eigenvectors in the non-sequential coupled basis (note that alternative bases also exist!) are:

$$\begin{aligned}|l_{12}, l_{34}, l, m\rangle \\ |1, 1, 2, m\rangle \\ |1, 1, 1, m\rangle \\ |1, 1, 0, m\rangle \\ |1, 0, 1, m\rangle \\ |0, 1, 1, m\rangle \\ |0, 0, 0, m\rangle\end{aligned}\tag{7.51}$$

As expected, the dimension of Hilbert space is preserved. This case also has one quintuplet, three triplets, and two singlets. ■

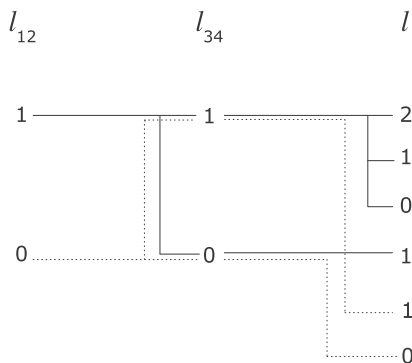


FIGURE 7.2 Addition of four $1/2$ angular momenta to build the non-sequential coupled basis.

7.4 Hilbert space expansion

We know that operators can be represented by matrices. Imagine that a certain system has two angular momenta $l_1 = 1/2$ and $l_2 = 1$. We immediately know that the Hilbert subspaces have different dimensions: 2×2 and 3×3 , respectively. How can we work with operators, which are matrices, with different dimensions? The solution is to expand the Hilbert space to a total dimension $\prod_{i=1}^N (2l_i + 1)$.

Objective The goal of this subsection is therefore to describe the methodology for matrix expansion of system operators.

To perform this task we can use the tensor products below:

$$\begin{aligned}
 \hat{L}_{1u} &= \hat{L}_{u(1)} \otimes \mathbb{1}_{(2)} \otimes \mathbb{1}_{(3)} \otimes \cdots \otimes \mathbb{1}_{(N)}, \\
 \hat{L}_{2u} &= \mathbb{1}_{(1)} \otimes \hat{L}_{u(2)} \otimes \mathbb{1}_{(3)} \otimes \cdots \otimes \mathbb{1}_{(N)}, \\
 &\vdots \\
 \hat{L}_{Nu} &= \mathbb{1}_{(1)} \otimes \mathbb{1}_{(2)} \otimes \mathbb{1}_{(3)} \otimes \cdots \otimes \hat{L}_{u(N)},
 \end{aligned} \tag{7.52}$$

where $u = (x, y, z)$. The subscript number outside the parentheses represents the n -th angular momentum, while the subscript number inside the parentheses represents the Hilbert subspace of the m -th momentum. For example, L_{2x} represents the component x of the second angular momentum, while $\mathbb{1}_{(2)}$ represents the identity operator in the subspace of the second moment. If this subspace refers to an angular momentum $l_2 = 2$,

then $\mathbb{1}_{(2)}$ has dimension 5×5 . After performing this tensor product,² for all N angular momenta and their three components (x , y , and z), each subspace achieves the correct dimension. Subsequently, algebraic operations can be carried out between the components of any angular momentum, as they will all share the same dimensionality.

As for the vectors, we must also expand them to the correct dimension of Hilbert space. In the case of N spins, this also requires a tensor product of N vectors:

$$\begin{aligned} |l_1, l_2, l_3, \dots, l_N, m_1, m_2, m_3, \dots, m_N\rangle = & \quad (7.53) \\ |l_1, m_1\rangle \otimes |l_2, m_2\rangle \otimes |l_3, m_3\rangle \otimes \dots \otimes |l_N, m_N\rangle = \\ \begin{pmatrix} \text{elements} \\ 2l_1 + 1 \end{pmatrix} \otimes \begin{pmatrix} \text{elements} \\ 2l_2 + 1 \end{pmatrix} \otimes \begin{pmatrix} \text{elements} \\ 2l_3 + 1 \end{pmatrix} \otimes \dots \otimes \begin{pmatrix} \text{elements} \\ 2l_N + 1 \end{pmatrix} \end{aligned}$$

■ Example 7.4

Let's consider the following operator:

$$\hat{H} = \vec{\hat{S}}_1 \cdot \vec{\hat{S}}_2 = \hat{S}_{1x}\hat{S}_{2x} + \hat{S}_{1y}\hat{S}_{2y} + \hat{S}_{1z}\hat{S}_{2z}, \quad (7.54)$$

where, in this example, we are considering $s_1 = 1/2$ and $s_2 = 1$. Note that the spin angular momentum follows the same algebra as the orbital angular momentum. We then have two Hilbert subspaces – one with dimension $2s_1 + 1 = 2$ and another with dimension $2s_2 + 1 = 3$, and consequently the dimension of the Hilbert space is given by $(2s_1 + 1)(2s_2 + 1) = 6$. As discussed above, we can obtain the components of the two spins in the three directions x , y , and z using Eq. (7.52). For example, the x -component of spin 1 is expressed as:

$$\hat{S}_{1x} = \hat{S}_{x(1)} \otimes \mathbb{1}_{(2)}, \quad (7.55)$$

where $\hat{S}_{x(1)}$ represents the spin operator $\hat{S}_x$ in subspace 1 (see Subsection 6.4), i.e. in dimension $2s_1 + 1 = 2$; and $\mathbb{1}_{(2)}$ the identity

$$a \otimes b = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \otimes \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = \begin{pmatrix} a_{11}b_{11} & a_{11}b_{12} & a_{12}b_{11} & a_{12}b_{12} \\ a_{11}b_{21} & a_{11}b_{22} & a_{12}b_{21} & a_{12}b_{22} \\ a_{21}b_{11} & a_{21}b_{12} & a_{22}b_{11} & a_{22}b_{12} \\ a_{21}b_{21} & a_{21}b_{22} & a_{22}b_{21} & a_{22}b_{22} \end{pmatrix}.$$

operator in subspace 2, i.e., in dimension $2s_2 + 1 = 3$. Thus:

$$\hat{S}_{1x} = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix}. \quad (7.56)$$

Similarly, the x -component of spin 2 reads as:

$$\hat{S}_{2x} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}. \quad (7.57)$$

The same procedure can be done for the other components of the two spins under analysis. We then leave this task to the reader as an exercise.

After completing these calculations, the algebraic operations in Eq. (7.54) are straightforward, as the subspaces are now in the correct dimensionality. In the proposed case, a standard matrix multiplication is carried out between the operators of the different spins and the same component (e.g., $\hat{S}_{1x}\hat{S}_{2x}$, etc.). Finally, by summing the three components of the operator, we obtain:

$$\hat{H} = \frac{\hbar^2}{2} \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sqrt{2} & 0 & 0 \\ 0 & 0 & -1 & 0 & \sqrt{2} & 0 \\ 0 & \sqrt{2} & 0 & -1 & 0 & 0 \\ 0 & 0 & \sqrt{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}. \quad (7.58)$$

As for the local basis vector, described by $|s_1, s_2, m_1, m_2\rangle = |s_1, m_1\rangle \otimes |s_2, m_2\rangle$, we can also obtain them and, for this purpose, we use Eq. (7.53). As an example, to avoid redundancy, we will present only one state:

$$|s_1, s_2, m_1, m_2\rangle = |s_1, m_1\rangle \otimes |s_2, m_2\rangle \quad (7.59)$$

$$|\frac{1}{2}, 1, \frac{1}{2}, 0\rangle = |\frac{1}{2}, \frac{1}{2}\rangle \otimes |1, 0\rangle \quad (7.60)$$

$$= \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}. \quad (7.61)$$

■

7.5 Local basis vs. coupled basis: Clebsch–Gordan coefficients

In the previous section, we discussed how to expand the Hilbert space for a system with N spins, i.e., how to obtain the operators under consideration in the correct dimension. Now, how to change the operators from the local basis to the coupled basis? To do this, we need to obtain the Clebsch–Gordan matrix – a matrix that contains all the coefficients that relate the vectors of the local basis to the vectors of the coupled basis. First, we will discuss the coupling of 2 angular momenta, and then the coupling of 3 angular momenta to obtain a recurrence relation for N angular momenta. However, the reader may be wondering: Why? One of the primary goals when working with operators is to determine their eigenvalues, which requires diagonalizing the operator. Depending on the operator under study, a common approach to achieve this is by transforming the basis from local to coupled.

Objective The goal of this section is therefore to obtain the Clebsch–Gordan matrix – a matrix containing all the coefficients that relate the local basis vector to the coupled basis vector.

7.5.1 Coupling of 2 angular momenta

Let's consider the total angular momentum of the system:

$$\vec{\hat{L}} = \vec{\hat{L}}_1 + \vec{\hat{L}}_2, \quad (7.62)$$

where each angular momentum has its corresponding vectors in their respective subspaces, i.e., $|l_1, m_1\rangle$ and $|l_2, m_2\rangle$, thus forming the local basis $|l_1, l_2, m_1, m_2\rangle$. The total angular momentum can be represented in the coupled basis $|l_1, l_2, l, m\rangle$.

Raising and lowering operators

If we consider $m_1 = m_{1\max} = l_1$ and $m_2 = m_{2\max} = l_2$, consequently (as discussed on page 258), $m = m_{\max} = l_1 + l_2$ and $l = l_{\max} = l_1 + l_2$. For this situation, there is only one possibility of states in the local and coupled basis, being a univocal relationship:

$$|l_1, l_2, l_{\max}, m_{\max}\rangle = |l_1, l_2, m_{1\max}, m_{2\max}\rangle \quad (7.63)$$

When applying the lowering operator $\hat{L}_- = \hat{L}_{1-} + \hat{L}_{2-}$ (Eq. (6.58)) $2l_{\max}$ times in the above equation, we can then find all vectors of $l_{\max}$ multiplet. To find the vectors of the $l_{\max} - 1$ multiplet, we use the orthogonality condition of the coupled basis vector:

$$\langle l_{\max}, m_{\max} - 1 | l_{\max} - 1, m_{\max} - 1 \rangle = 0 \quad (7.64)$$

After that, it is possible to apply the lowering operators again. This procedure can be performed for all multiplets of the system, from $l = l_{\max}$ to $l = l_{\min}$. The numbers associating one vector to another are known as Clebsch–Gordan coefficients.

★ **Rudolf Alfred Clebsch (1833–1872):** Born in Königsberg, Prussia. He collaborated with Paul Gordan to introduce the Clebsch–Gordan coefficients for spherical harmonics, indispensable in quantum mechanics.

Paul Albert Gordan (1837–1912): Born in Wroclaw, Prussia. He was a mathematician renowned for his contributions to invariant theory, the development of Clebsch–Gordan coefficients, and the formulation of Gordan’s lemma.

■ Example 7.5

Let’s consider, as in Example 7.1, two angular momenta $l_1 = l_2 = 1/2$. We then have the local and coupled bases, both known (Eqs. (7.41) and (7.42)); and now we will determine the relationship between these vectors. We start with the condition of Eq. (7.63), where $m_1 = m_{1\max} = l_1$ and $m_2 = m_{2\max} = l_2$, and, consequently, $m = m_{\max} = l_1 + l_2$ and $l = l_{\max} = l_1 + l_2$. Thus, using the reduced notation where l_1 and l_2 are omitted, it is possible to write:

$$|1, +1\rangle = |+, +\rangle. \quad (7.65)$$

Remember: $+$ represents $+1/2$, and $-$ represents $-1/2$. Then, let’s apply the lowering operator $\hat{L}_- = \hat{L}_{1-} + \hat{L}_{2-}$ in the above equation:

$$\hat{L}_- |1, +1\rangle = (\hat{L}_{1-} + \hat{L}_{2-}) |+, +\rangle. \quad (7.66)$$

Using Eq. (6.58), we find:

$$|1, 0\rangle = \frac{1}{\sqrt{2}}(|+, -\rangle + |-, +\rangle). \quad (7.67)$$

We then apply the operator $\hat{L}_- = \hat{L}_{1-} + \hat{L}_{2-}$ again:

$$\hat{L}_-|1, 0\rangle = \frac{1}{\sqrt{2}}(\hat{L}_{1-} + \hat{L}_{2-})(|+, -\rangle + |-, +\rangle), \quad (7.68)$$

which, after a few steps and with the help of Eq. (6.58) results in:

$$|1, -1\rangle = |-, -\rangle. \quad (7.69)$$

We need to obtain the vector for the singlet $l = 0$ and, to do this, we use the orthogonality condition (Eq. (7.64)):

$$\langle 1, 0|0, 0\rangle = 0. \quad (7.70)$$

We know $\langle 1, 0|$ from Eq. (7.67), but what about $|0, 0\rangle$? From Eq. (7.34), we know that $m = m_1 + m_2$ and, consequently, $|0, 0\rangle$ is associated only with the states $|+, -\rangle$ and $|-, +\rangle$. Therefore, we can write:

$$|0, 0\rangle = a|+, -\rangle + b|-, +\rangle. \quad (7.71)$$

Therefore, the above equation under the orthogonality condition (7.70) leads to:

$$\frac{1}{\sqrt{2}}(\langle +, -| + \langle -, +|)(a|+, -\rangle + b|-, +\rangle) = 0. \quad (7.72)$$

As a solution to the above equation, we then have $b = -a$. Furthermore, from the normalization condition $|a|^2 + |b|^2 = 1$ we determine that $a = 1/\sqrt{2}$ and, finally:

$$|0, 0\rangle = \frac{1}{\sqrt{2}}(|+, -\rangle - |-, +\rangle). \quad (7.73)$$

■

Wigner 3j symbol and Racah formula

There is, however, another way to obtain the relationship between the local and coupled bases. To do this, we consider the completeness relation of the vectors of the local basis:

$$\sum_{m_1} \sum_{m_2} |l_1, l_2, m_1, m_2\rangle \langle l_1, l_2, m_1, m_2| = \mathbb{1}, \quad (7.74)$$

where $\mathbb{1}$ is the identity operator. Applying this relation only by the right side of the equality $|l_1, l_2, l, m\rangle = |l_1, l_2, l, m\rangle$ we obtain:

$$|l_1, l_2, l, m\rangle = \sum_{m_1} \sum_{m_2} \langle l_1, l_2, m_1, m_2 | l_1, l_2, l, m \rangle |l_1, l_2, m_1, m_2\rangle \quad (7.75)$$

where

$$\langle l_1, l_2, m_1, m_2 | l_1, l_2, l, m \rangle = (-1)^{l_1 - l_2 + m} (2l + 1)^{1/2} \begin{pmatrix} l_1 & l_2 & l \\ m_1 & m_2 & -m \end{pmatrix} \quad (7.76)$$

are the Clebsch–Gordan coefficients, which, in turn, can be expressed in terms of the Wigner $3j$ symbol (2×3 object on the right – see Box 7.3 for more details). These coefficients are always real and form a matrix with the dimension of the Hilbert space of the total system. This matrix takes the system from the local basis to the coupled basis, which is precisely the objective of this section. It is also worth mentioning that this matrix is unitary (see Application 3.A).

Example 7.6 makes clearer the systematic calculation of the Clebsch–Gordan coefficients and the construction of the change of basis matrix.

Box 7.3 Detailed description of Wigner $3j$ symbol and Racah formula

To obtain the Clebsch–Gordan coefficients, we have to solve the Wigner $3j$ symbol, which, in turn, can be solved using the Racah formula:

$$\begin{pmatrix} a & b & c \\ A & B & C \end{pmatrix} = (-1)^{a-b-C} Q^{1/2} (\Delta_{abc})^{1/2} \sum_{n=n_{\min}}^{n_{\max}} \frac{(-1)^n}{f(n)}, \quad (7.77)$$

where

$$Q = (a + A)!(a - A)!(b + B)!(b - B)!(c + C)!(c - C)!, \quad (7.78)$$

$$\Delta_{abc} = \frac{(a + b - c)!(b + c - a)!(c + a - b)!}{(a + b + c + 1)!}, \quad (7.79)$$

$$f(n) = n!(n - n_1)!(n - n_2)!(n_3 - n)!(n_4 - n)!(n_5 - n)!, \quad (7.80)$$

$$n_1 = -c + b - A, \quad (7.81)$$

$$n_2 = -c + a + B, \quad (7.82)$$

$$n_3 = a + b - c, \quad (7.83)$$

$$n_4 = b + B, \quad (7.84)$$

$$n_5 = a - A, \quad (7.85)$$

$$n_{\min} = \max\{0; \max\{n_1, n_2\}\} \quad \text{and} \quad n_{\max} = \min\{n_3; \min\{n_4, n_5\}\}. \quad (7.86)$$

The Racah formula is equal to zero if it does not satisfy the conditions below:

$$A + B + C = 0 \quad \text{and} \quad |a - b| \leq c \leq a + b, \quad (7.87)$$

and, consequently, the Clebsch–Gordan coefficients are also zero if they do not satisfy these conditions, that is:

$$m = m_1 + m_2 \quad \text{and} \quad |l_1 - l_2| \leq l \leq l_1 + l_2. \quad (7.88)$$

These two conditions above ratify what we had already determined (Eqs. (7.34) and (7.31)).

■ Example 7.6

In this example, we calculate the Clebsch–Gordan coefficients and consequently the change of basis matrix – from the local to the coupled basis. Let's consider two $1/2$ angular momenta, that is $l_1 = l_2 = 1/2$. Since:

$$|l_1 - l_2| \leq l \leq l_1 + l_2, \quad (7.89)$$

thus: $0 \leq l \leq 1$. From these information, the vectors of the two bases (local and coupled) can be constructed:

$$\text{local basis } (|m_1, m_2\rangle) : \begin{Bmatrix} |+, +\rangle \\ |+, -\rangle \\ |-, +\rangle \\ |-, -\rangle \end{Bmatrix} \quad \text{coupled basis } (|l, m\rangle) : \begin{Bmatrix} |1, 1\rangle \\ |1, 0\rangle \\ |1, -1\rangle \\ |0, 0\rangle \end{Bmatrix}$$

See Example 7.1 for more details. The Clebsch–Gordan coefficients measure the degree of correlation between the vectors of each basis. Since we have four vectors in each basis, we then have 16 coefficients, relating all the vectors of the local basis with all the vectors of the coupled basis. From Eq. (7.75) it is possible to write:

$$\begin{aligned} |1, 1\rangle &= \langle +, + | 1, 1 \rangle |+, +\rangle + \langle +, - | 1, 1 \rangle |+, -\rangle + \langle -, + | 1, 1 \rangle |-, +\rangle \\ &\quad + \langle -, - | 1, 1 \rangle |-, -\rangle \end{aligned} \quad (7.90)$$

$$\begin{aligned} |1, 0\rangle &= \langle +, + | 1, 0 \rangle |+, +\rangle + \langle +, - | 1, 0 \rangle |+, -\rangle + \langle -, + | 1, 0 \rangle |-, +\rangle \\ &\quad + \langle -, - | 1, 0 \rangle |-, -\rangle \end{aligned} \quad (7.91)$$

$$\begin{aligned} |1, -1\rangle &= \langle +, + | 1, -1 \rangle |+, +\rangle + \langle +, - | 1, -1 \rangle |+, -\rangle + \langle -, + | 1, -1 \rangle |-, +\rangle \\ &\quad + \langle -, - | 1, -1 \rangle |-, -\rangle \end{aligned} \quad (7.92)$$

$$\begin{aligned} |0, 0\rangle &= \langle +, + | 0, 0 \rangle |+, +\rangle + \langle +, - | 0, 0 \rangle |+, -\rangle + \langle -, + | 0, 0 \rangle |-, +\rangle \\ &\quad + \langle -, - | 0, 0 \rangle |-, -\rangle \end{aligned} \quad (7.93)$$

and, consequently:

$$|l, m\rangle = \hat{U}|m_1, m_2\rangle \quad (7.94)$$

where

$$\hat{U} = \begin{pmatrix} \langle +, +|1, 1\rangle & \langle +, -|1, 1\rangle & \langle -, +|1, 1\rangle & \langle -, -|1, 1\rangle \\ \langle +, +|1, 0\rangle & \langle +, -|1, 0\rangle & \langle -, +|1, 0\rangle & \langle -, -|1, 0\rangle \\ \langle +, +|1, -1\rangle & \langle +, -|1, -1\rangle & \langle -, +|1, -1\rangle & \langle -, -|1, -1\rangle \\ \langle +, +|0, 0\rangle & \langle +, -|0, 0\rangle & \langle -, +|0, 0\rangle & \langle -, -|0, 0\rangle \end{pmatrix} \quad (7.95)$$

is the change of basis matrix, containing the Clebsch–Gordan elements.

The null elements can be easily identified, as they consist of all elements that fail to satisfy the relation expressed in Eq. (7.34). The non-null elements are determined using Eq. (7.76), derived from the Racah formula (Eq. (7.77)). It is worth noting that the equation under consideration is challenging to solve manually; however, a straightforward computational routine allows for rapid execution.³ Thus, the change of basis matrix reads as:

$$\hat{U} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} & 0 \end{pmatrix}. \quad (7.96)$$

From this matrix $\hat{U}$, we then have the relationship between the vectors of the two bases:

$$|1, 1\rangle = |+, +\rangle \quad (7.97)$$

$$|1, 0\rangle = \frac{1}{\sqrt{2}}(|+, -\rangle + |-, +\rangle)$$

$$|1, -1\rangle = |-, -\rangle$$

$$|0, 0\rangle = \frac{1}{\sqrt{2}}(|+, -\rangle - |-, +\rangle).$$

Note that this result is the same (and could not be different) as that obtained by applying the operator $\hat{L}_-$ successively (Eqs. (7.65), (7.67), (7.69), and (7.73)).

We can also obtain the inverse relationship, as the matrix $\hat{U}$ is unitary, that is, $\hat{U}^{-1} = \hat{U}^\dagger$ (see Eq. (3.A.2)). Thus, from Eq. (7.94), it is possible to write:

³A viable option for computational calculation is to use the Wolfram Alpha website. On this platform, type ‘Clebsch–Gordan calculator’.

$$\begin{aligned}
 |m_1, m_2\rangle &= \hat{U}^{-1} |l, m\rangle \\
 &= \hat{U}^\dagger |l, m\rangle,
 \end{aligned}
 \tag{7.98}$$

where

$$\hat{U}^\dagger = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 0 & 1 & 0 \end{pmatrix}
 \tag{7.99}$$

and, consequently:

$$\begin{aligned}
 |+, +\rangle &= |1, 1\rangle \\
 |+, -\rangle &= \frac{1}{\sqrt{2}} (|1, 0\rangle + |0, 0\rangle) \\
 |-, +\rangle &= \frac{1}{\sqrt{2}} (|1, 0\rangle - |0, 0\rangle) \\
 |-, -\rangle &= |1, -1\rangle.
 \end{aligned}
 \tag{7.100}$$

■

7.5.2 Coupling of 3 angular momenta

To continue this discussion and finally obtain a recurrence relation for the coupling of N angular momenta, we need to discuss first the coupling of 3 momenta: $\vec{L}_1$, $\vec{L}_2$, and $\vec{L}_3$. The local basis is quite intuitive and easy to set up: $|l_1, l_2, l_3, m_1, m_2, m_3\rangle$. As for the coupled basis, it can be sequential or non-sequential, but for reasons of simplicity, we will consider in this and next subsections the sequential coupled basis $|(l_1, l_2, [l_{12}]), l_3, l, m\rangle$. The parentheses (and brackets) always contain three angular momenta, the third being the sum of the first two. For this case, $l = l_{13}$ and $m = m_{13}$.

To obtain the elements of the change of basis matrix, we first consider the coupling between the first and second angular momentum (similar to what was done before):

$$|l_1, l_2, l_{12}, m_{12}\rangle = \sum_{m_1} \sum_{m_2} \langle l_1, l_2, m_1, m_2 | l_1, l_2, l_{12}, m_{12} \rangle |l_1, l_2, m_1, m_2\rangle
 \tag{7.101}$$

and then coupling the above result with the third angular momentum:

$$\begin{aligned}
 &|(l_1, l_2, [l_{12}]), l_3, l, m\rangle \\
 &= \sum_{m_{12}} \sum_{m_3} \langle l_1, l_2, l_{12}, l_3, m_{12}, m_3 | l_1, l_2, l_{12}, l_3, l, m \rangle |l_1, l_2, l_{12}, l_3, m_{12}, m_3\rangle.
 \end{aligned}
 \tag{7.102}$$

However, we know that:

$$|l_1, l_2, l_{12}, l_3, m_{12}, m_3\rangle = |l_1, l_2, l_{12}, m_{12}\rangle \otimes |l_3, m_3\rangle \quad (7.103)$$

and taking Eq. (7.101) into Eq. (7.102), we find:

$$\begin{aligned} & |(l_1, l_2, [l_{12}], l_3, l, m) \\ &= \sum_{m_{12}} \sum_{m_3} \langle l_1, l_2, l_{12}, l_3, m_{12}, m_3 | l_1, l_2, l_{12}, l_3, l, m \rangle \\ & \quad \times \left\{ \sum_{m_1} \sum_{m_2} \langle l_1, l_2, m_1, m_2 | l_1, l_2, l_{12}, m_{12} \rangle |l_1, l_2, m_1, m_2\rangle \right\} \otimes |l_3, m_3\rangle. \end{aligned} \quad (7.104)$$

As the final result we obtain:

$$\begin{aligned} & |(l_1, l_2, [l_{12}], l_3, l, m) \\ &= \sum_{m_{12}} \sum_{m_1} \sum_{m_2} \sum_{m_3} \langle l_1, l_2, l_{12}, l_3, m_{12}, m_3 | l_1, l_2, l_{12}, l_3, l, m \rangle \\ & \quad \langle l_1, l_2, m_1, m_2 | l_1, l_2, l_{12}, m_{12} \rangle |l_1, l_2, l_3, m_1, m_2, m_3\rangle. \end{aligned} \quad (7.105)$$

We then observe that in the case of the coupling of 3 angular momenta, the local basis vectors are related to the coupled basis vectors through the product of two Clebsch–Gordan coefficients. The first coefficient refers to the coupling of the first angular momentum with the second one, forming a $\hat{L}_{12}$ vector that, in turn, couples with the third angular moment, producing the second Clebsch–Gordan coefficient.

7.5.3 Recurrence relation for N angular momenta

Now it is easier to obtain a recurrence relation for the change of basis of a system with N angular momenta. As we know, the coupling of 2 angular momenta is given by the relation:

$$|(l_1, l_2, l_{12}), m_{12}\rangle = \hat{U}^{12} |l_1, l_2, m_1, m_2\rangle. \quad (7.106)$$

In the sequence, the coupling of 3 and 4 angular momenta can be obtained, respectively, through the relations below:

$$\begin{aligned} |(l_1, l_2, [l_{12}], l_3, l_{13}), m_{13}\rangle &= \hat{U}^{13} |(l_1, l_2, l_{12}), m_{12}\rangle \otimes |l_3, m_3\rangle \\ &= \hat{U}^{13} \hat{U}^{12} |l_1, l_2, l_3, m_1, m_2, m_3\rangle \end{aligned} \quad (7.107)$$

and

$$\begin{aligned} |(l_1, l_2, [l_{12}], l_3, (l_{13}), l_4, l_{14}), m_{14}\rangle &= \hat{U}^{14} |(l_1, l_2, [l_{12}], l_3, l_{13}), m_{13}\rangle \otimes |l_4, m_4\rangle \\ &= \hat{U}^{14} \hat{U}^{13} \hat{U}^{12} |l_1, l_2, l_3, l_4, m_1, m_2, m_3, m_4\rangle. \end{aligned} \quad (7.108)$$

Finally, the coupling of N angular momenta occurs through the recurrence relation below:

$$\begin{aligned} & |(l_1, l_2, [l_{12}], l_3, [l_{13}], l_4, l_{14}), \dots, [l_{1(n-1)}], l_n, l, m\rangle \\ &= \hat{U}^{1n} |(l_1, l_2, [l_{12}], l_3, [l_{13}], l_4, l_{14}), \dots, [l_{1(n-2)}], l_{n-1}, l_{1(n-1)}\rangle \otimes |l_n, m_n\rangle \\ &= \hat{U}^{1n} \dots \hat{U}^{14} \hat{U}^{13} \hat{U}^{12} |l_1, l_2, l_3, l_4, \dots, l_n, m_1, m_2, m_3, m_4, \dots, m_n\rangle \end{aligned} \quad (7.109)$$

which, as can be seen, is the product of $N - 1$ Clebsch–Gordan coefficients. Above, $l = l_{1n}$ and $m = m_{1n}$. Below is the description of these coefficients:

$$\hat{U}^{12} = \langle l_1, l_2, m_1, m_2 | l_1, l_2, l_{12}, m_{12} \rangle, \quad (7.110)$$

$$\hat{U}^{13} = \langle l_{12}, l_3, m_{12}, m_3 | l_{12}, l_3, l_{13}, m_{13} \rangle, \quad (7.111)$$

$$\hat{U}^{14} = \langle l_{13}, l_4, m_{13}, m_4 | l_{13}, l_4, l_{14}, m_{14} \rangle, \quad (7.112)$$

$$\vdots \quad (7.113)$$

$$\hat{U}^{1n} = \langle l_{1(n-1)}, l_n, m_{1(n-1)}, m_n | l_{1(n-1)}, l_n, l, m \rangle. \quad (7.114)$$

■ Example 7.7

Let's use the result of Example 7.2, where we coupled 4 angular momenta $l_1 = l_2 = l_3 = l_4 = 1/2$. We know the possible vectors, both in the local basis and in the sequential coupled basis:

$$\begin{array}{cc} \text{local basis} & \text{coupled basis} \\ (|m_1, m_2, m_3, m_4\rangle) : & (|l_{12}, l_{13}, l, m\rangle) : \end{array} \left\{ \begin{array}{l} |+, +, +, +\rangle \\ |+, +, +, -\rangle \\ |+, +, -, +\rangle \\ |+, +, -, -\rangle \\ |+, -, +, +\rangle \\ |+, -, +, -\rangle \\ |+, -, -, +\rangle \\ |+, -, -, -\rangle \\ |-, +, +, +\rangle \\ |-, +, +, -\rangle \\ |-, +, -, +\rangle \\ |-, +, -, -\rangle \\ |-, -, +, +\rangle \\ |-, -, +, -\rangle \\ |-, -, -, +\rangle \\ |-, -, -, -\rangle \end{array} \right\} \left\{ \begin{array}{l} |1, 3/2, 2, 2\rangle \\ |1, 3/2, 2, 1\rangle \\ |1, 3/2, 2, 0\rangle \\ |1, 3/2, 2, -1\rangle \\ |1, 3/2, 2, -2\rangle \\ |1, 3/2, 1, 1\rangle \\ |1, 3/2, 1, 0\rangle \\ |1, 3/2, 1, -1\rangle \\ |1, 1/2, 1, 1\rangle \\ |1, 1/2, 1, 0\rangle \\ |1, 1/2, 1, -1\rangle \\ |1, 1/2, 0, 0\rangle \\ |0, 1/2, 1, 1\rangle \\ |0, 1/2, 1, 0\rangle \\ |0, 1/2, 1, -1\rangle \\ |0, 1/2, 0, 0\rangle \end{array} \right\}$$

In this case, we have a change of basis matrix $\hat{U}$ with dimension 16×16 and therefore 256 elements – too many to be calculated in this example. Thus, let's calculate two elements: one non-zero and one zero. Starting with the element:

$$\langle +, +, +, - | 1, 3/2, 1, 1 \rangle \quad (7.115)$$

expressed in notation:

$$\langle m_1, m_2, m_3, m_4 | l_{12}, l_{13}, l, m \rangle. \quad (7.116)$$

As we have done before, '+' means $+1/2$ and '-' means $-1/2$. This element is in the sixth row and second column of the matrix $\hat{U}$. To calculate this element, we need to calculate three Clebsch–Gordan coefficients:

$$\hat{U}^{12} \hat{U}^{13} \hat{U}^{14} \quad (7.117)$$

where, as described in Eqs. (7.110)–(7.114):

$$\hat{U}^{12} = \hat{U}^{12}(l_1, l_2, l_{12}, m_1, m_2, m_{12}), \quad (7.118)$$

$$\hat{U}^{13} = \hat{U}^{13}(l_{12}, l_3, l_{13}, m_{12}, m_3, m_{13}),$$

$$\hat{U}^{14} = \hat{U}^{14}(l_{13}, l_4, l, m_{13}, m_4, m).$$

We know that: $l_1 = 1/2$, $l_2 = 1/2$, $l_{12} = 1$, $m_1 = +1/2$, $m_2 = +1/2$, $l_3 = 1/2$, $l_{13} = 3/2$, $m_3 = +1/2$, $l_4 = 1/2$, $l = 1$, $m_4 = -1/2$, and $m = 1$, but we do not yet know m_{12} and m_{13} . Nevertheless, we know that $l_{12} = 1$ and $l_{13} = 3/2$ and thus we determine $m_{12} = 1, 0, -1$ and $m_{13} = 3/2, 1/2, 0, -1/2, -3/2$. Therefore, we can calculate the proposed matrix element:

$$\langle +, +, +, - | 1, 3/2, 1, 1 \rangle = \quad (7.119)$$

$$\sum_{m_{12}} \sum_{m_{13}} \hat{U}^{12}(m_{12}) \hat{U}^{13}(m_{12}, m_{13}) \hat{U}^{14}(m_{13}) =$$

$$\hat{U}^{12}(m_{12} = 1) \hat{U}^{13}(m_{12} = 1, m_{13} = 3/2) \hat{U}^{14}(m_{13} = 3/2) +$$

$$\hat{U}^{12}(m_{12} = 1) \hat{U}^{13}(m_{12} = 1, m_{13} = 1/2) \hat{U}^{14}(m_{13} = 1/2) +$$

$\vdots$

$$\hat{U}^{12}(m_{12} = -1) \hat{U}^{13}(m_{12} = -1, m_{13} = -3/2) \hat{U}^{14}(m_{13} = -3/2).$$

Note that in the expression above, we explicitly highlight the dependence of $\hat{U}^{1i}$ on m_{12} and m_{13} . However, in practice, these terms also depend on other quantum numbers, as indicated in Eq. (7.118). However, many of the terms in the above equation are zero, as they do not satisfy Eq. (7.34), which in turn should be rewritten as:

$$m_{12} = m_1 + m_2 \quad (7.120)$$

$$m_{13} = m_{12} + m_3.$$

Therefore, we can determine $m_{12} = 1$, and, consequently, $m_{13} = 3/2$. If m_{12} and m_{13} assume other values, the Clebsch–Gordan coefficients are zero. Thus, the matrix element analyzed is summarized as:

$$\langle +, +, +, - | 1, 3/2, 1, 1 \rangle = \quad (7.121)$$

$$\hat{U}^{12}(m_{12} = 1) \hat{U}^{13}(m_{12} = 1, m_{13} = 3/2) \hat{U}^{14}(m_{13} = 3/2) = 1.1. \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}.$$

To complete this example, let's calculate another matrix element: $\langle +, +, +, + | 1, 3/2, 1, 1 \rangle$, where, in this case, only $m_4 = 1/2$ is different from the element calculated above. Again, from Eq. (7.34) (and, consequently, from Eq. (7.120)), we obtain: $m_{12} = 1$ and $m_{13} = 3/2$. Thus:

$$\hat{U}^{12}(1/2, 1/2, 1, 1/2, 1/2, 1) = \hat{U}^{13}(1, 1/2, 3/2, 1, 1/2, 3/2) = 1, \quad (7.122)$$

as calculated above. However:

$$\hat{U}^{14}(3/2, 1/2, 1, 3/2, 1/2, 1) = 0, \quad (7.123)$$

since this coefficient does not satisfy the condition $m_{14} = m_{13} + m_4$. Finally:

$$\langle +, +, +, + | 1, 3/2, 1, 1 \rangle = 0. \quad (7.124)$$

It is important to emphasize that the numerical value of the Clebsch–Gordan coefficients described above (Eq. (7.122), for example), were obtained using the Racah formula (Eq. (7.77)) and, due to the size of the aforementioned formula, a computational routine was used. ■

7.6 Basis change of an operator

In the previous section, we determined the change of basis matrix, from local to coupled (and vice versa). As we mentioned before, this matrix allows us to (i) write the coupled basis vector in terms of the local basis vector (and vice versa) (ii) change the basis of an operator. Why change the basis? In many cases, the operator is not diagonal in the local basis but becomes diagonal in the coupled basis. Diagonalizing the operator is essential because it allows us to access its eigenvalues, which are critical for understanding its properties.

Objective The goal of this section is to use the change of basis matrix and exemplify the methodology of how to take a given operator from the local basis to the coupled basis.

To change the basis of the operator, we highly recommend revisiting Appendix 3.A, where Eq. (3.A.6), presented below, is derived:

$$\hat{A}_d = \hat{U} \hat{A}_a \hat{U}^\dagger, \quad (7.125)$$

where $\hat{A}_d$ is a certain operator after the change of basis, $\hat{U}$ is the change of basis matrix, $\hat{U}^\dagger$ is the conjugate transpose of the change of basis matrix, and finally $\hat{A}_a$ is the operator before the change of basis. As has been done throughout this book, let's look at an example, which will make the ideas clearer.

■ Example 7.8

First, let's consider Example 7.4, where we studied the operator:

$$\hat{H} = \vec{\hat{S}}_1 \cdot \vec{\hat{S}}_2 = \hat{S}_{1x} \hat{S}_{2x} + \hat{S}_{1y} \hat{S}_{2y} + \hat{S}_{1z} \hat{S}_{2z}. \quad (7.126)$$

In this example, we are considering $s_1 = s_2 = 1/2$, such that the above operator, after the due expansion of the Hilbert space (see Exercise 7.12), reads as:

$$\hat{H} = \frac{\hbar^2}{4} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 2 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (7.127)$$

Note that the above operator, written in the local basis, is not diagonal. The change of basis matrix, for this case, was obtained in Example 7.6 (Eq. (7.96)), and therefore the change of basis can be written as:

$$\hat{H}_d = \hat{U} \hat{H}_a \hat{U}^\dagger, \quad (7.128)$$

where $\hat{H}_d$ is given by Eq. (7.127). Thus:

$$\begin{aligned} \hat{H}_d &= \frac{\hbar^2}{4} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 2 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \\ &\quad \times \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 0 & 1 & 0 \end{pmatrix} \end{aligned} \quad (7.129)$$

and, consequently:

$$\hat{H}_d = \frac{\hbar^2}{4} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -3 \end{pmatrix}. \quad (7.130)$$

The operator $\hat{H}$ is diagonal, featuring three degenerate states with an eigenvalue of $1/4$ and one non-degenerate state with an eigenvalue of $-3/4$. It is important to highlight that the change of basis from local to coupled does not necessarily ensure the diagonalization of the operator in question. ■

Application 7.A Magnetic moment and the Zeeman effect

The Hamiltonian describing the Zeeman effect comes directly from the Dirac equation (see Chapter 13), more precisely from Eqs. (13.163) and (13.164):

$$\hat{H}_Z = \frac{\mu_B}{\hbar} \left(2\vec{\hat{S}} + \vec{\hat{L}} \right) \cdot \vec{B} \quad (7.A.1)$$

where, for convenience (as done in Section 7.1), we will make the angular momenta dimensionless and, for this reason, we will omit the pre-factor $1/\hbar$. Thus, the Zeeman interaction can be rewritten as:

$$\hat{H}_Z = \mu_B \left(2\vec{\hat{S}} + \vec{\hat{L}} \right) \cdot \vec{B} \quad (7.A.2)$$

Therefore, note that the Zeeman effect promotes an interaction of angular momenta (spin $\vec{\hat{S}}$ and orbital $\vec{\hat{L}}$), with the magnetic field $\vec{B}$. Above, $\mu_B = e\hbar/2m$ is the Bohr magneton, where e/m is the charge-to-mass ratio of the electron. First, we will consider the purely spin contribution of the Hamiltonian:

$$\hat{H}_Z^{(S)} = 2\mu_B \vec{\hat{S}} \cdot \vec{B} = -\vec{\hat{\mu}}_S \cdot \vec{B} \quad (7.A.3)$$

where

$$\vec{\hat{\mu}}_S = -g_S \mu_B \vec{\hat{S}} \quad (7.A.4)$$

and $g_S = 2$ is the gyromagnetic factor. Similar to the above case, we will consider the purely orbital contribution:

$$\hat{H}_Z^{(L)} = \mu_B \vec{\hat{L}} \cdot \vec{B} = -\vec{\hat{\mu}}_L \cdot \vec{B} \quad (7.A.5)$$

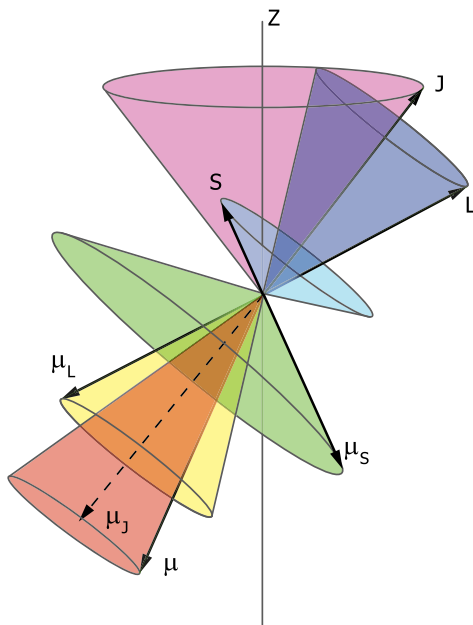


FIGURE 7.3 Orbital $\vec{L}$ and spin $\vec{S}$ angular momenta added together to obtain the total angular momentum $\vec{J}$. These vectors have magnitude $\sqrt{u(u+1)}\hbar$, where u can be either l , s or j . Adapted from reference [17].

where

$$\vec{\mu}_L = -g_L \mu_B \vec{\hat{L}} \quad (7.A.6)$$

and $g_L = 1$. The total Hamiltonian can be written as:

$$\hat{\mathcal{H}} = -\vec{\hat{\mu}} \cdot \vec{B}, \quad (7.A.7)$$

where

$$\hat{\mathcal{H}} = \hat{\mathcal{H}}^{(L)} + \hat{\mathcal{H}}^{(S)}. \quad (7.A.8)$$

By comparing Eqs. (7.A.7) and (7.A.8), we obtain:

$$\vec{\hat{\mu}} = -\mu_B \left(\vec{\hat{L}} + 2\vec{\hat{S}} \right). \quad (7.A.9)$$

See Fig. 7.3 to better understand this scenario. Note that the total magnetic moment $\vec{\hat{\mu}}$ is not parallel to the total angular momentum $\vec{\hat{J}} = \vec{\hat{L}} + \vec{\hat{S}}$, due to the different gyromagnetic factors of the spin and orbital momenta. There is an important point to highlight: In the previous chapter, we concluded that the expected values of the x - and y -components of the angular

momentum are zero, i.e., $\langle \hat{L}_x \rangle = \langle \hat{L}_y \rangle = 0$. This result arises because, if we know the magnitude of the angular momentum and its projection along the z -axis, the uncertainty principle prevents us from gaining additional information about its components in the x - and y -directions. Likewise, in the present case, the expected value of the angular momentum component $\vec{\mu}$ perpendicular to the direction of the total angular momentum $\vec{J}$ is zero, and therefore only $\vec{\mu}_J$ (antiparallel to $\vec{J}$) contributes to the magnetic energy. Therefore, we multiply both sides of Eq. (7.A.9) by the unity vector $\vec{J}/\hat{J}$, where we will obtain the projection of the total magnetic momentum $\vec{\mu}$ along the vector $\vec{J}$. Thus:

$$\vec{\mu}_J = \left(\frac{\vec{J}}{\hat{J}} \cdot \vec{\mu} \right) \frac{\vec{J}}{\hat{J}} = -\mu_B \left(\frac{\vec{L} \cdot \vec{J}}{\hat{J}} + 2 \frac{\vec{S} \cdot \vec{J}}{\hat{J}} \right) \frac{\vec{J}}{\hat{J}}. \quad (7.A.10)$$

Considering that $\vec{J} = \vec{L} + \vec{S}$, then:

$$\hat{S}^2 = (\vec{J} - \vec{L}) \cdot (\vec{J} - \vec{L}) = \hat{J}^2 + \hat{L}^2 - 2\vec{L} \cdot \vec{J}, \quad (7.A.11)$$

$$\hat{L}^2 = (\vec{J} - \vec{S}) \cdot (\vec{J} - \vec{S}) = \hat{J}^2 + \hat{S}^2 - 2\vec{S} \cdot \vec{J}. \quad (7.A.12)$$

Taking the above equations into Eq. (7.A.10), we obtain:

$$\vec{\mu}_J = -\frac{\mu_B}{2\hat{J}^2} (3\hat{J}^2 - \hat{L}^2 + \hat{S}^2) \vec{J}. \quad (7.A.13)$$

By analogy with Eqs. (7.A.4) and (7.A.6), it is possible to write:

$$\vec{\mu}_J = -g_J \mu_B \frac{\vec{J}}{\hat{J}}, \quad (7.A.14)$$

and, comparing the above equation with Eq. (7.A.13), we achieve at:

$$g_J = \left[\frac{3\hat{J}^2 - \hat{L}^2 + \hat{S}^2}{2\hat{J}^2} \right]. \quad (7.A.15)$$

Considering the coupled basis, written with the following vectors $\{|s, l, j, m_j\rangle\}$, the following expectation value can be written:

$$g_J = \langle s, l, j, m_j | \left[\frac{3\hat{J}^2 - \hat{L}^2 + \hat{S}^2}{2\hat{J}^2} \right] | s, l, j, m_j \rangle \quad (7.A.16)$$

which implies in:

$$g \equiv g_J = 1 + \frac{j(j+1) - l(l+1) + s(s+1)}{2j(j+1)} \quad (7.A.17)$$

where g is the well-known Landé factor. Finally, the Zeeman Hamiltonian can be rewritten as:

$$\hat{H}_Z = -\vec{\mu}_J \cdot \vec{B} = g\mu_B \vec{J} \cdot \vec{B} \quad (7.A.18)$$

where $\vec{\mu}_J$ is given by Eq. (7.A.14). It is worth mentioning that in this development we are considering the Landé factor as a scalar, a number. However, for more advanced studies, depending on the material, this factor can be anisotropic, being represented in the equations above by a tensor. The detailed description of how to represent the Landé factor as a tensor is beyond the scope of this book, and this information is of a complementary nature [17].

★ **Pieter Zeeman (1865–1943):** Born in Zonnemaire, in the province of Zeeland, Netherlands. He was awarded the 1902 Nobel Prize in Physics for his research on the influence of magnetism on radiation phenomena.

Alfred Landé (1888–1976): Born in Elberfeld, Germany. He had Arnold Sommerfeld as his PhD advisor, with the Landé factor, discussed above, being one of his main contributions.

Application 7.B Angular momentum of atoms and the Hund's rules

Atoms have several electrons, and, in this case, how can we add their angular momenta or, at least, have some idea? Complete atomic shells do not contribute to the total angular momentum because their net contribution is precisely zero. Therefore, only the electrons in the incomplete shells contribute to the total angular momentum of the atom. In the case of light atoms (with an atomic number less than 75), the spin angular momentum of each electron couples to produce an effective spin:

$$\vec{\hat{S}} = \sum_i \vec{\hat{S}}_i, \quad (7.B.1)$$

that is, the effective spin angular momentum is the sum of the spin angular momentum of each electron. Similarly, the orbital angular momenta also couple, resulting in an effective orbital angular momentum:

$$\vec{\hat{L}} = \sum_i \vec{\hat{L}}_i. \quad (7.B.2)$$

These angular momenta, $\vec{\hat{S}}$ and $\vec{\hat{L}}$, interact through spin-orbit coupling to produce the total effective angular momentum:

$$\hat{\mathcal{H}} = \zeta \vec{\hat{S}} \cdot \vec{\hat{L}} \quad (7.B.3)$$

where ζ is the spin-orbit parameter. For heavy atoms (atomic number greater than 75), each electron couples its spin and orbital momentum to produce its own total angular momentum. These individual total angular momenta then couple to produce an effective total angular momentum:

$$\vec{J} = \sum_i \vec{J}_i. \quad (7.B.4)$$

This last case is known as $j - j$ coupling.

Based on the discussion above, we are able to introduce the Hund's rules:

Hund's first rule: effective spin momentum According to the Pauli exclusion principle, no more than two electrons can occupy the same orbital (the two electrons must have opposing spins). To minimize electron repulsion, the orbitals are filled one by one and then, after all are filled, the first orbital receives a second electron (with opposite spin, 'down' in this case). This means that the effective spin must be at its maximum value. This is Hund's first rule.

Hund's second rule: effective orbital momentum For a given effective spin, there are several possible values for the effective orbital momentum, but the main question is: which configuration minimizes the energy of the system? The answer also lies in the electronic repulsion: if all the electrons precess in the same direction,⁴ this minimizes the chance of these electrons meeting each other. Therefore, for the energy to be minimum, the effective orbital momentum must be maximum. This is Hund's second rule.

Hund's third rule: total momentum This last Hund's rule is not as intuitive as the previous ones, and originates from the spin-orbit coupling (Eq. (7.B.3)). The ζ parameter is positive when the shell is less than half filled, and negative when the shell is more than half filled. Therefore, for the first case, $\vec{S}$ must be antiparallel to $\vec{L}$ to minimize the energy; and thus $\vec{J} = \vec{L} - \vec{S}$ is the ground state. For the second case, $\vec{S}$ must be parallel to $\vec{L}$; and then $\vec{J} = \vec{L} + \vec{S}$ is the ground state.

Consider the simple case of a rare-earth ion: Sm^{3+} . The electronic configuration of this element is:

$$Sm : [Xe]4f^6 6s^2 \Rightarrow Sm^{3+} : [Xe]4f^5. \quad (7.B.5)$$

Let's recall the well-known orbital filling diagram (see Table 7.1). Hund's first rule states that the effective spin momentum is the sum of the individual momentum. Thus, for this example: $s = 1/2 + 1/2 + 1/2 + 1/2 + 1/2 = 5/2$. Note that s is a quantum number that measures the magnitude of

⁴A semi-classical approach, such as the Bohr model.

TABLE 7.1 Diagram to fill the orbitals of Sm^{3+} . The numbers represent m_j (eigenvalue of $\hat{J}_z$), while each arrow represents a single electron (with spin ‘up’ or ‘down’).

+3	+2	+1	0	−1	−2	−3
↑	↑	↑	↑	↑		

TABLE 7.2 Spectroscopic notation for orbital momentum l : S-Sharp, P-Principal, D-Diffuse, F-Fundamental. The following values of l , that is greater than 3, follow the alphabetical order: G, H, I, and so forth.

l	0	1	2	3	4	5	6	...
X	S	P	D	F	G	H	I	...

the vector $\vec{\hat{S}}$, i.e., $\hat{S}^2|s, l, j, m_j\rangle = s(s+1)\hbar^2|s, l, j, m_j\rangle$. From Hund’s second rule, and by analogy with Hund’s first rule, we then have $l = +3 + 2 + 1 + 0 - 1 = +5$. Finally, using Hund’s third rule, it is possible to determine the value of j , considering that, for this case, the f shell is less than half filled (f shell holds a maximum of 14 electrons). Thus, the ground state is the multiplet $j = |s - l| = 5/2$. It is important to emphasize that the system has several multiplets, from $|s - l|$ (ground state) to $s + l$. The multiplets are: $j = 5/2$ (ground state), $7/2$, $11/2$, $13/2$ and $15/2$.

These angular momenta can be written using Russell–Saunders notation, known as the *Term Symbol*:

$$^{2s+1}X_j \tag{7.B.6}$$

where X represents the effective orbital quantum number l in spectroscopic notation (see Table 7.2 for a quick review). For the case of Sm^{3+} , the ground state can be written as $^6H_{5/2}$.

★ **Friedrich Hund (1896–1997):** Born in Karlsruhe, Germany. Known for his contributions to the study of atoms and molecules. However, his main achievements of physics and chemistry are the Hund’s rules, discussed above.

Appendix 7.A Exercises

Conceptual exercises

- Exercise 7.1.** Why should we expand the Hilbert space?
- Exercise 7.2.** Describe the physical mechanism that justifies Hund’s first and second rules.
- Exercise 7.3.** What condition does the spin-orbit coupling parameter impose on Hund’s third rule?

Text-based exercises

Exercise 7.4. Consider $\vec{L} = \vec{L}_1 + \vec{L}_2$ - see Section 7.2.

1. Calculate $[\hat{L}_z, \hat{L}_x]$.
2. Calculate $[\hat{L}_y, \hat{L}_z]$.

Exercise 7.5. Consider the operator of the Example 7.4:

$$\hat{H} = \vec{\hat{S}}_1 \cdot \vec{\hat{S}}_2 = \hat{S}_{1x}\hat{S}_{2x} + \hat{S}_{1y}\hat{S}_{2y} + \hat{S}_{1z}\hat{S}_{2z}. \quad (7.A.1)$$

where $s_1 = 1/2$ and $s_2 = 1$. Obtain the spin operators $\hat{S}_{1y}$, $\hat{S}_{2y}$, $\hat{S}_{1z}$, and $\hat{S}_{2z}$ in the matrix form.

Practice exercises

Exercise 7.6. Consider $\vec{L} = \vec{L}_1 + \vec{L}_2$ - see Section 7.2.

1. Calculate $[\hat{L}^2, \hat{L}_{1z}]$.
2. Calculate $[\hat{L}^2, \hat{L}_{2z}]$.
3. Do the observables $\{\hat{L}_1^2, \hat{L}_2^2, \hat{L}^2, \hat{L}_{1z}\}$ share the same basis vector? Justify your answer.

Exercise 7.7. 1. What is the condition for the Clebsch–Gordan coefficients to be non-zero?

2. Deduce this condition.

Exercise 7.8. Can Clebsch–Gordan coefficients be complex numbers?

Exercise 7.9. Consider two spins: $s_1 = 1$ and $s_2 = 2$. Formulate the vectors of the local basis and the coupled basis.

Exercise 7.10. Consider three spins: $s_1 = 1$, $s_2 = 2$ and $s_3 = 1$. Formulate the sequential and non-sequential coupled basis. For the non-sequential basis, consider the sum of the first spin with the third spin; and then combine the result with the second spin. Verify the dimension of the Hilbert space.

Exercise 7.11. Consider four spins: $s_1 = 1$ and $s_2 = s_3 = s_4 = 1/2$. Formulate the sequential and non-sequential coupled basis. For the non-sequential basis, consider the sum of the first spin with the second spin; then the sum of the third spin with the fourth spin; and finally, the sum of the two previous results. Verify the dimension of the Hilbert space.

Exercise 7.12. Write the operator $\hat{H} = \vec{\hat{S}}_1 \cdot \vec{\hat{S}}_2$ in the matrix form. Consider $s_1 = s_2 = 1/2$. Obtain the vector $|+, -\rangle$.

Exercise 7.13. Write the operator $\hat{H} = \hat{S}_z^2$ in the matrix form. Consider $\hat{S}_z = \hat{S}_{1z} + \hat{S}_{2z} + \hat{S}_{3z}$ and $s_1 = s_2 = s_3 = 1/2$.

Exercise 7.14. Consider two angular momenta: $l_1 = 1/2$ and $l_2 = 1$.

1. Write the equations that relate the coupled basis to the local basis, analogous to the Eqs. (7.90)–(7.93).
2. Write the matrix $\hat{U}$, analogous to the Eq. (7.95).
3. Identify the non-zero elements.

Exercise 7.15. Consider two angular momenta: $l_1 = l_2 = 1$.

1. Write the vectors for the local and coupled bases.
2. Apply the operator $\hat{L}_- = \hat{L}_{1-} + \hat{L}_{2-}$ successively and obtain the relationship between the vectors of the local and coupled bases.
3. Formulate the Clebsch–Gordan matrix, that is, the basis change matrix.

Development exercises

Exercise 7.16. Obtain the eigenvalues of the operator $\hat{H} = \vec{\hat{S}}_1 \cdot \vec{\hat{S}}_2$ using only algebraic operations. Identify the multiplets. Consider $s_1 = s_2 = 1/2$.

Exercise 7.17. Obtain the eigenvalues of the operator $\hat{H} = \hat{L}_z^2$ using only algebraic operations. Identify the states degeneracy. Consider $l_1 = l_2 = 1$.

Advanced exercises

Exercise 7.18. Consider the Hamiltonian:

$$\hat{\mathcal{H}} = -J_1(\vec{\hat{S}}_1 \cdot \vec{\hat{S}}_2) - J_2(\vec{\hat{S}}_1 \cdot \vec{\hat{S}}_3 + \vec{\hat{S}}_2 \cdot \vec{\hat{S}}_3) \quad (7.A.2)$$

Obtain its eigenvalues and identify its multiplets. Consider $s_1 = s_2 = 1/2$ and $s_3 = 1$. Also consider that J_1 and J_2 are constants.

Exercise 7.19. Consider four ions with spins $s = 1/2$ interacting through a Heisenberg-type Hamiltonian $\hat{\mathcal{H}} = -J\vec{\hat{S}}_i \cdot \vec{\hat{S}}_j$ (see Fig. 7.4). Dashed lines indicate an interaction of intensity J_1 , while the solid lines indicate an interaction J_2 . Obtain the following:

1. The total Hamiltonian of the system.
2. The basis that diagonalizes the Hamiltonian.
3. The vectors of this basis.
4. The energy spectra of the system. Identify the multiplets.

Hint: it is not necessary to solve the problem using matrices.

Exercise 7.20. Consider the rare-earth ion Pr^{3+} , with electronic configuration $4f^2$. Write, in Russell–Saunders notation, known as the *Term Symbol*, the first excited state of the ion.

Exercise 7.21. Consider a Cr^{4+} ion with $3d^2$ electron configuration.

1. Write in spectroscopic notation all the multiplets of the ion, identifying the ground state and the last excited state.

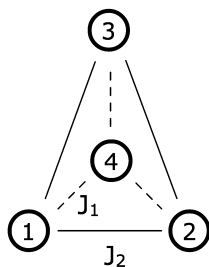


FIGURE 7.4 Spatial configuration of spins and corresponding magnetic interactions. This figure refers to the Exercise 7.19.

2. The ion has an anisotropic Hamiltonian of the form $\frac{\beta}{2}(\hat{J}_x^2 + \hat{J}_y^2)$. Write the total Hamiltonian of the system. Consider $\beta > 0$ and $\vec{J}$ as the total angular momentum.
3. In which basis is the previous Hamiltonian diagonal?
4. Write an expression for the energy of the system.
5. Considering the multiplets, write the energy spectrum.
6. Does the anisotropic term increase or decrease the energy of the multiplets?

Exercise 7.22. The trivalent Terbium ion has an electronic configuration of $4f^8$, and the energy difference between the ground state and the first excited state is 3.000 Kk_B .

1. Write the multiplets of this ion in Russell–Saunders notation, identifying the ground state and the last excited state.
2. Write an expression for the energy of the system.
3. Considering the multiplets, write the energy spectrum.
4. Obtain the energy value of the ground state (Kk_B), considering that the last excited state is at the zero of energy.

Exercise 7.23. Consider five spins: $s_1 = s_2 = s_3 = s_4 = s_5 = 1/2$. The Hamiltonian describing the interaction between these spins is given by:

$$\hat{\mathcal{H}} = -J_1(\vec{\hat{S}}_1 \cdot \vec{\hat{S}}_2 + \vec{\hat{S}}_2 \cdot \vec{\hat{S}}_3) - J_2(\vec{\hat{S}}_A \cdot \vec{\hat{S}}_B) - J_3(\vec{\hat{S}}_4 \cdot \vec{\hat{S}}_5) \quad (7.A.3)$$

where $\vec{\hat{S}}_A = \vec{\hat{S}}_1 + \vec{\hat{S}}_2 + \vec{\hat{S}}_3$ and $\vec{\hat{S}}_B = \vec{\hat{S}}_4 + \vec{\hat{S}}_5$.

1. Choose a coupled basis that diagonalizes this Hamiltonian and write the system's energy.
2. Obtain the eigenvalues of this Hamiltonian and identify the multiplets.